

10.7.26

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Question

- ① The focal chord to $y^2 = 16x$ is tangent to $(x - 6)^2 + y^2 = 2$, then the possible values of the slope of this chord are: (2003)
- ① $-1, 1$
 - ② $-2, 2$
 - ③ $-2, -\frac{1}{2}$
 - ④ $2, -\frac{1}{2}$

Theoretical Solution

Let us solve the given equation theoretically and then verify the solution computationally

According to the question,

The parabola is

$$y^2 = 16x. \quad (1)$$

In quadratic form, this can be written as

$$\mathbf{x}^T \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} + 2 \begin{pmatrix} -8 \\ 0 \end{pmatrix}^T \mathbf{x} = 0. \quad (2)$$

The circle is

$$(x - 6)^2 + y^2 = 2 \Rightarrow x^2 + y^2 - 12x + 34 = 0, \quad (3)$$

which can be expressed as

$$\mathbf{x}^T \mathbf{I} \mathbf{x} + 2 \begin{pmatrix} -6 \\ 0 \end{pmatrix}^T \mathbf{x} + 34 = 0. \quad (4)$$

Theoretical Solution

Thus,

$$\mathbf{V} = \mathbf{I}, \quad \mathbf{u} = \begin{pmatrix} -6 \\ 0 \end{pmatrix}, \quad f = 34. \quad (5)$$

The focus of the parabola is

$$\mathbf{F} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}. \quad (6)$$

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Theoretical Solution

tangency condition

$$\mathbf{m}^T(\mathbf{q}\mathbf{v} + \mathbf{u}) = 0 \quad (7)$$

$$\mathbf{q} \text{ is point of contact} \quad (8)$$

$$(9)$$

equation of tangent

$$(\mathbf{q}\mathbf{v} + \mathbf{u})^T \mathbf{x} + \mathbf{u}^T \mathbf{q} + f = 0 \quad (10)$$

$$\mathbf{m}^T(\mathbf{q}\mathbf{v} + \mathbf{u}) = 0 \quad (11)$$

$$(\mathbf{q}\mathbf{v} + \mathbf{u})^T \mathbf{F} + \mathbf{u}^T \mathbf{q} + f = 0 \quad (12)$$

$$\mathbf{q}^T \mathbf{v} \mathbf{q} + 2 \begin{pmatrix} -6 \\ 0 \end{pmatrix}^T \mathbf{q} + 34 = 0. \quad (13)$$

by solving above equations we get

$$m = 1, -1$$

Plot

